
Resource-Aware Finite Representation of Continuous Geometric Objects

Fidelity, Structural Validity, Task Adequacy, and Certified Resource Complexity

Pearl Bipin Pulickal

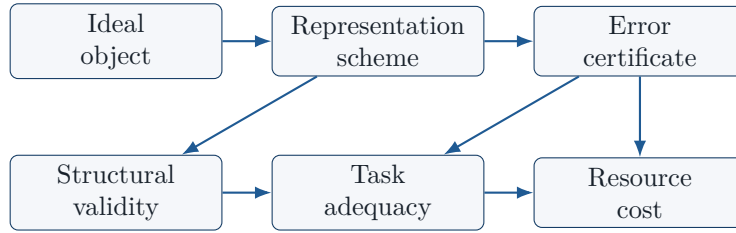
Independent Researcher

Electronics Engineering Graduate, National Institute of Technology Goa, India

September 22, 2026

ANALYTICAL PREPRINT

A mathematical framework for determining how much finite representation is sufficient under explicit fidelity, validity, task, and resource constraints.



Abstract

A continuous mathematical object and any finite computational or physical representation of it are different mathematical entities. This paper studies the resources required to construct a finite representation that is sufficiently faithful and structurally valid for a prescribed purpose. The motivating example is the circle: mathematically, it contains a continuum of ideal points, whereas a finite system must use a finite number of samples, grid cells, encoded coordinates, arithmetic operations, or other finite resources. The central question is therefore not how many mathematical points a circle contains, nor how many digits of π exist, but how much finite representation is sufficient under an explicit tolerance, structural requirement, representation scheme, and cost model.

The paper develops a representation-level framework in which a continuous object X is mapped through a specified representation scheme to a finite object. Geometric fidelity and structural validity are kept as separate constraints, while a usage contract records the operations and resource costs that the representation must support. A certified representation complexity is defined as the infimum cost, within a specified representation class and

cost model, among representations for which a rigorous error certificate and the required structural conditions are satisfied. A certificate-preserving conversion relation is included as supporting machinery for comparing finite representation schemes. The principal quantitative result is a sharp mixed discretization–finite-bit allocation theorem: for the certified model $A/N^2 + D2^{-b} \leq \varepsilon$ with normalized storage cost Nb , the optimal resource law has the form $\varepsilon^{-1/2}[\log(1/\varepsilon) + \log \log(1/\varepsilon) + O(1)]$, and the continuous relaxation admits an explicit asymptotic allocation of the error budget. A general resource-allocation theorem is also proved for separable power-law error and cost models, with the distinction between continuous relaxations and genuinely discrete optimization made explicit.

The circle serves as an exactly analyzable laboratory. For the regular inscribed N -gon, the Hausdorff error is exactly $r(1 - \cos(\pi/N))$, giving the exact least admissible sample count within that representation family. The sequence of regular polygons converges to the circle in Hausdorff distance, with error of order N^{-2} . A refined centered scaling of the regular polygon is also analyzed. Spatial quantization by nearest-grid vertex rounding and certified numerical perturbation are then added through a rigorous triangle-inequality decomposition. Exact closure is treated as a structural property rather than as a consequence of small geometric error. A decision-stability result shows how the arithmetic precision required for a discrete decision depends on the decision margin, not on a universal number of decimal digits of π .

The paper further distinguishes mathematical points from finite-resolution physical marks, derives a simple packing bound for separated samples, and introduces an abstract task-adequacy condition. Under a Lipschitz stability assumption for a task functional, a task margin can be converted into a sufficient geometric tolerance. Simple representation-complexity models illustrate that different representations of the same circle can have different resource-accuracy scaling laws, while the sharp mixed discretization–finite-bit theorem makes one such law quantitatively explicit. The paper treats polygonal approximation, digital geometry, task-based quantization, rate–distortion–perception, adaptive precision, and mixed-precision analysis as established neighboring areas and does not claim to invent those fields. Its quantitative contribution is intentionally narrow: a rigorous, auditable representation-level framework together with the explicit asymptotic solution of the stated circle-derived mixed resource model.

Keywords. continuous-to-discrete representation; representation complexity; polygonal approximation; Hausdorff distance; quantization; finite precision; certified computation; decision stability; task adequacy; resource allocation; digital geometry.

Core message: finite representation is meaningful only relative to an explicit fidelity criterion, structural or task requirement, usage contract, and resource cost.

Contents

1	Introduction	5
2	Scope, positioning, and contribution	6
2.1	Established neighboring areas	6
2.2	Narrow scope adopted here	6
2.3	Claims and non-claims	6
2.4	Objects, representations, and physical realizations	7
2.4.1	The mathematical circle has infinitely many ideal points	7
2.4.2	Finite separated physical samples	8
2.4.3	The regular polygon sequence	8

2.5	Representation contracts and certified complexity	8
2.5.1	Representation schemes	8
2.5.2	Usage and access contracts	9
2.5.3	Geometric error and structural validity	9
2.5.4	Certified versus actual error	9
3	Exact geometry of the circle model	10
3.1	Uniform angular sampling	10
3.2	Chord length and perimeter	11
3.3	Exact Hausdorff error	11
3.4	Convergence to the circle	11
3.5	Minimum sample count for a prescribed Hausdorff tolerance	12
3.6	A refined centered regular-polygon model	12
3.7	Chord-length resolution is a different requirement	13
4	Certified finite-representation pipeline	13
4.1	Nearest-grid quantization	14
4.2	A numerical perturbation layer	15
4.3	Structural validity and exact closure	15
4.3.1	Exact closure by construction	15
4.3.2	Closure under independent perturbation	15
5	Decision stability and numerical precision	16
5.1	Decision margins	16
5.2	Application to a circle step count	16
5.3	Geometric tolerance and integer decisions	17
5.4	What the precision of π actually controls	17
5.4.1	What irrationality does not imply	17
5.4.2	When the precision of π matters	17
6	Resource-allocation theory	17
6.1	Separable power-law models	17
6.2	Continuous versus discrete optimization	18
6.3	The circle as a resource-allocation model	19
6.3.1	Two geometric resources	19
6.3.2	Unique minimizer of the relaxed toy model	19
6.3.3	Error budgets	20
6.3.4	Certified bottlenecks	20
6.4	Certificate-compatible representation conversion	20
6.5	Representation-dependent resource complexity	21
6.5.1	Polygonal sample-count complexity	21
6.5.2	Full-grid raster complexity under an explicit contract	22
6.5.3	Finite-bit coordinates for polygonal samples	22
6.5.4	Representation competition	25

6.6	Further consequences and feasibility	25
6.6.1	Monotonicity	25
6.6.2	Physical-resolution feasibility window	25
6.6.3	A representation-complexity transfer principle	25
7	Purpose-specific adequacy	26
7.1	Why geometric accuracy need not be the final specification	26
7.2	A transfer theorem from geometric error to task adequacy	26
7.3	Diminishing returns	27
7.4	Static versus multiresolution representations	27
8	Literature context and status of the contribution	27
8.1	Polygonal approximation and quantization	27
8.2	Digital geometry	28
8.3	Robust predicates and precision	28
8.4	Mixed precision and resource allocation	28
8.5	Perception and task-aware representation	28
8.6	Limitations and claim boundaries	28
8.6.1	The quantization model is intentionally simple	28
8.6.2	The numerical perturbation parameter is abstract	29
8.6.3	The cost model is not universal	29
8.6.4	Certified complexity may exceed actual complexity	29
8.6.5	Representation-class dependence	29
8.6.6	Task adequacy requires task information	29
8.7	Research status and contribution statement	29
9	Open problems and extensions	30
10	Conclusion	30
A	Elementary inequalities and asymptotics	31
B	Numerical illustration	31
C	Methodological scope	32

1 Introduction

A continuous mathematical object and a finite representation of that object should not be conflated. A Euclidean circle is an ideal set in $\mathbb{R}^2$ containing a continuum of points. A computational or physical system may instead use a finite ordered list of samples, a polygonal boundary, a raster, a collection of grid cells, quantized coordinates, an integer recurrence, or a finite-precision numerical representation. The transition from the first object to the second is therefore a representation problem rather than merely a question of evaluating one constant more accurately.

The motivating question of this paper may be phrased informally as follows:

How much finite precision and representation resource are enough to represent a continuous object for a specified purpose?

The word “enough” is essential. Increasing resolution generally improves a representation only up to the requirements of the task, while the corresponding resource cost may continue to increase. A display, a measurement device, a geometric computation, and a high-consequence engineering procedure may therefore justify different representation requirements for the same underlying object. This paper does not attempt to model human vision or any particular application empirically. Instead, it develops a mathematical framework in which the required tolerance or task specification is stated explicitly.

The circle is a particularly useful laboratory because its geometry is exact. Let

$$\Gamma_r := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = r^2\}, \quad r > 0.$$

There are many finite representations of Γ_r . A regular polygon can be generated by uniform angular sampling. A grid representation can use lattice cells or quantized vertices. An integer circle construction can avoid an explicit high-precision representation of π . These representations are mathematically legitimate but have different resource requirements and different numerical failure modes.

The paper is organized around five distinctions. First, the mathematical object is separated from its finite representation. Second, geometric fidelity is separated from structural validity such as exact closure. Third, actual error is separated from a certified upper bound. Fourth, the representation scheme and its usage contract are treated as part of the problem rather than hidden background assumptions. Fifth, continuous optimization is separated from the final discrete optimization over integers or finite formats.

The resulting question is

Among admissible finite representations, what is the infimum of the specified resource cost at a prescribed fidelity and validity level?

$$\mathcal{C}_{\mathfrak{R}, \mathcal{K}}^{\text{cert}}(X, \varepsilon) = \inf_{\mathcal{R} \in \mathfrak{R}} \mathcal{C}_{\mathcal{R}, \mathcal{K}}^{\text{cert}}(X, \varepsilon) \tag{1}$$

The contribution is deliberately stated conservatively. Polygonal approximation, grid digitization, robust predicates, task-based quantization, rate-distortion-perception, and mixed-precision resource allocation all have substantial prior literature. The paper does not claim those ideas individually. Instead, it develops a common representation-level language, proves exact and certifiable results for the circle, and makes the dependence of finite resource requirements on representation, access contract, tolerance, and decision margin explicit.

2 Scope, positioning, and contribution

Structural guide

The organizing framework. The paper separates five ingredients that are often conflated in informal discussions of “precision”: the ideal object, the finite representation scheme, the fidelity criterion, the structural or task requirements imposed on the representation, and the resource cost used to compare admissible representations. None of these ingredients is intrinsic to the abstract object alone; the resulting complexity is meaningful only relative to the stated representation contract.

2.1 Established neighboring areas

The problem studied here intersects several mature literatures. Polygonal approximation treats both minimum error at fixed complexity and minimum complexity at fixed error; the classical formulations of Imai and Iri and later algorithmic work provide foundational references [1, 2]. Closed-curve approximation has also been studied explicitly [3]. Joint polygonal approximation and vertex quantization has been treated as a lossy vector-data coding problem, including minimum bit-rate at a prescribed deviation [4].

Digital geometry studies the relation between Euclidean shapes and finite grids, including Hausdorff convergence and topological properties of digitizations [5, 6]. Robust computational geometry studies adaptive precision and certified discrete predicates [7]. Mixed-precision numerical analysis studies the interaction between discretization error and quantization or rounding error, including progressive-precision strategies [9, 10].

More broadly, rate–distortion and rate–distortion–perception theories study the allocation of finite representation resources under fidelity and perceptual constraints [13]. Task-based quantization deliberately designs finite representations around a downstream task rather than only around reconstruction fidelity [14]. Computational resource-allocation frameworks likewise optimize several resolution or discretization variables jointly for a specified prediction task [12]. These precedents are important because they rule out a generic novelty claim that the present paper “discovers” resource-aware approximation or task-aware representation as broad principles.

2.2 Narrow scope adopted here

The focus is narrower: deterministic continuous geometric objects, explicit finite geometric representations, certified geometric error, structural validity, finite numerical perturbation, and representation-dependent resource cost. The principal test case is the circle. The framework is intentionally abstract enough to admit storage, sample count, arithmetic work, latency, energy, or a weighted combination as a cost, but no single cost is treated as canonical.

2.3 Claims and non-claims

The paper claims the following mathematical results within the explicitly defined models:

- (i) exact Hausdorff error and exact sample-count thresholds for the regular inscribed circle model;
- (ii) Hausdorff convergence of the regular polygon sequence to the circle;
- (iii) rigorous quantization and numerical-perturbation bounds for a staged representation

- pipeline;
- (iv) a general representation-complexity definition that distinguishes actual feasibility from certified feasibility;
 - (v) a certificate-preserving conversion bound for comparing finite representation schemes, used as supporting framework machinery;
 - (vi) a sharp asymptotic mixed discretization–finite-bit allocation theorem for the explicit circle-derived model $A/N^2 + D2^{-b} \leq \varepsilon$ with normalized cost Nb ;
 - (vii) a continuous error-budget allocation theorem for a separable power-law class;
 - (viii) exact structural closure by endpoint identity and a quantitative closure-defect bound under independent perturbations; and
 - (ix) a task-adequacy transfer result under an explicit Lipschitz stability assumption.

The paper does not claim to establish a universal cost model, a universal human-perception threshold, a universal optimal representation over all conceivable encodings, or a new theory of rate–distortion, task-based quantization, digital geometry, or robust geometric computation. Statements about broader representations, human usefulness, energy, medical systems, zoom-dependent rendering, and other applications are therefore kept as motivation, conditional interpretation, or open problems unless a precise mathematical model is supplied.

2.4 Objects, representations, and physical realizations

2.4.1 The mathematical circle has infinitely many ideal points

The statement that a circle contains infinitely many points is mathematically correct. In fact, the map

$$[0, 2\pi) \ni \theta \longmapsto (r \cos \theta, r \sin \theta) \in \Gamma_r$$

is a bijection, so Γ_r has the same cardinality as a real interval. These mathematical points, however, are ideal zero-dimensional elements; they do not consume positive physical area merely by being points of the set.

This should be separated sharply from a physical representation. A mark made by a pen, a pixel, a laser spot, a stored coordinate, or a detector sample has finite resolution or finite resource cost. Such a finite mark is not literally a mathematical point.

Remark 2.1 There is therefore no contradiction between an object having infinitely many mathematical points and having finite length. The interval $[0, 1]$ is an elementary example: it contains infinitely many points but has finite length 1. The phrase “ $N \rightarrow \infty$ ” in an approximation theorem denotes a limit of a sequence of finite representations; it does not describe an endlessly growing physical circle.

Remark 2.2 (The physical “growing infinity” intuition) There is, however, a useful physical intuition behind the contrary impression. Suppose one tries to construct a circle by placing distinct physical marks around a fixed finite circumference, with each mark occupying nonzero physical extent or being separated from the next by a positive minimum resolvable distance. Then one cannot keep adding distinguishable marks indefinitely while keeping the circumference and resolution fixed. Once the available spatial resolution has been exhausted, an additional mark must either overlap an existing mark, fail to create a distinguishable new sample, or require a change in the physical scale or representation. In that physical model, an attempt to keep increasing the number of distinguishable marks beyond the available capacity cannot be implemented by simply “adding infinity”; the finite resource constraint is reached first. This

does *not* mean that the mathematical circle grows when its set of ideal points is infinite. It means only that a physical realization cannot assign a distinct finite-sized resource to every mathematical point.

2.4.2 Finite separated physical samples

The physical intuition becomes mathematically precise once a finite resolution is imposed.

Proposition 2.3 (Packing bound for separated samples). *Let $S \subset \Gamma_r$ be a finite set of distinct sample points listed in cyclic order. Suppose the arclength distance between every pair of consecutive samples is at least $\delta > 0$. Then*

$$|S| \delta \leq 2\pi r,$$

and hence

$$|S| \leq \left\lfloor \frac{2\pi r}{\delta} \right\rfloor.$$

Proof. The consecutive samples partition the circumference into $|S|$ circular arcs. Each such arc has length at least δ , while their total length is $2\pi r$. Therefore $|S|\delta \leq 2\pi r$. Since $|S|$ is an integer, the stated bound follows. $\square$

The proposition does not assign a finite number of points to the mathematical circle. It states that a physical or computational representation with a prescribed minimum separation admits only finitely many distinguishable samples.

2.4.3 The regular polygon sequence

A nondegenerate closed polygonal boundary requires at least three distinct vertices. With three vertices one obtains a triangle; with four a quadrilateral, and the regular cases give the familiar sequence of the equilateral triangle, square, regular pentagon, regular hexagon, and so forth.

For $N \geq 3$, let P_N denote the boundary of the regular N -gon whose vertices lie on Γ_r and are equally spaced in angle. The statement that the polygon “becomes” a circle should be interpreted as convergence:

$$P_N \longrightarrow \Gamma_r \quad (N \rightarrow \infty),$$

not as equality for any finite N .

2.5 Representation contracts and certified complexity

2.5.1 Representation schemes

Definition 2.4 (Representation scheme) Let $\mathcal{X}$ be a class of continuous objects. A representation scheme $\mathcal{R}$ consists of a finite-dimensional parameter space $\mathcal{Q}_{\mathcal{R}}$ and a map

$$\mathcal{R} : (X, q) \longmapsto D_{\mathcal{R}}(X; q), \quad q \in \mathcal{Q}_{\mathcal{R}},$$

where $D_{\mathcal{R}}(X; q)$ is a finite or discretely encoded object intended to represent X . The parameter space may be continuous, while each admissible parameter value specifies a finite or discretely encoded representation.

The same continuous object may have several representation schemes. Examples include explicit point sets, polygonal boundaries, rasterizations, implicit integer constructions, or quantized coordinates.

2.5.2 Usage and access contracts

A finite encoding cannot be compared meaningfully by storage alone when different encodings support different operations. To make resource comparisons explicit, we associate with a representation scheme a *usage contract* $\mathcal{K}$. The contract specifies the admissible downstream operations, which intermediate objects must be materialized, and which resource components are counted by the cost model.

Thus a cost should be written conceptually as

$$C_{\mathcal{R},\mathcal{K}}(q),$$

not merely as a property of the abstract mathematical object. This prevents an implicit representation such as $x^2 + y^2 = r^2$ from being declared universally optimal merely because the equation occupies few symbols when the actual task requires a large set of spatial queries or explicit samples.

Remark 2.5 Any comparison between representation families is therefore relative to a stated usage contract and cost model. This is a structural limitation of the theory, not a defect: “optimal” has meaning only relative to what the representation must support and what resources are being counted.

2.5.3 Geometric error and structural validity

When A and B are compact subsets of a metric space, the Hausdorff distance is

$$d_H(A, B) := \max \left\{ \sup_{a \in A} \text{dist}(a, B), \sup_{b \in B} \text{dist}(b, A) \right\}.$$

Other metrics may be appropriate for other tasks, including directed Hausdorff distance, Fréchet distance, area error, radial error, normal deviation, or a task-specific functional. Hausdorff distance is used in the circle model because it gives exact and interpretable statements.

Definition 2.6 (Structural validity) Let $P_{\mathcal{K}}(X, D)$ be a Boolean predicate encoding a structural requirement imposed by the usage contract. The representation D is structurally valid when

$$P_{\mathcal{K}}(X, D) = 1.$$

Examples include exact closure, connectedness, prescribed orientation, non-self-intersection, or correctness of a discrete predicate.

Geometric fidelity and structural validity are logically independent. A very accurate approximation can still fail exact closure, while a structurally valid representation may be geometrically too coarse.

2.5.4 Certified versus actual error

Let $E_{\mathcal{R}}(X; q)$ denote the actual geometric error and let $\hat{E}_{\mathcal{R}}(X; q)$ be a certified upper bound satisfying

$$E_{\mathcal{R}}(X; q) \leq \hat{E}_{\mathcal{R}}(X; q).$$

The distinction is essential because an optimization based on a conservative certificate can require more resources than the true optimum.

Definition 2.7 (Certified representation complexity) For tolerance $\varepsilon > 0$, define

$$C_{\mathcal{R},\mathcal{K}}^{\text{cert}}(X, \varepsilon) := \inf \left\{ C_{\mathcal{R},\mathcal{K}}(q) : \hat{E}_{\mathcal{R}}(X; q) \leq \varepsilon, P_{\mathcal{K}}(X, D_{\mathcal{R}}(X; q)) = 1 \right\},$$

with the convention $\inf \emptyset = +\infty$.

The corresponding actual representation complexity is

$$\mathcal{C}_{\mathcal{R},\mathcal{K}}^{\text{act}}(X, \varepsilon) := \inf \{C_{\mathcal{R},\mathcal{K}}(q) : E_{\mathcal{R}}(X; q) \leq \varepsilon, P_{\mathcal{K}}(X, D_{\mathcal{R}}(X; q)) = 1\}.$$

Since certified feasibility implies actual feasibility,

$$\boxed{\mathcal{C}_{\mathcal{R},\mathcal{K}}^{\text{act}}(X, \varepsilon) \leq \mathcal{C}_{\mathcal{R},\mathcal{K}}^{\text{cert}}(X, \varepsilon).}$$

The equality need not hold.

Proposition 2.8 (Monotonicity in the tolerance). *If $0 < \varepsilon_1 \leq \varepsilon_2$, then*

$$\mathcal{C}_{\mathcal{R},\mathcal{K}}^{\text{cert}}(X, \varepsilon_1) \geq \mathcal{C}_{\mathcal{R},\mathcal{K}}^{\text{cert}}(X, \varepsilon_2).$$

Proof. Every representation certified at tolerance ε_1 is also certified at tolerance ε_2 . Thus the feasible set for ε_1 is contained in the feasible set for ε_2 , and the infimum over the smaller set cannot be lower. $\square$

For a representation class $\mathfrak{R}$ with compatible usage and cost conventions, define

$$\mathcal{C}_{\mathfrak{R},\mathcal{K}}^{\text{cert}}(X, \varepsilon) := \inf_{\mathcal{R} \in \mathfrak{R}} \mathcal{C}_{\mathcal{R},\mathcal{K}}^{\text{cert}}(X, \varepsilon).$$

This quantity is an infimum within the stated class; it need not be attained without additional compactness, discreteness, or coercivity assumptions. It is not an intrinsic complexity of X independent of representation choices.

3 Exact geometry of the circle model

3.1 Uniform angular sampling

Let

$$\Gamma_r := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = r^2\}, \quad r > 0.$$

For $N \geq 3$ define

$$\theta_k = \frac{2\pi k}{N}, \quad p_k = (r \cos \theta_k, r \sin \theta_k), \quad k = 0, \dots, N-1,$$

and let

$$P_N := [p_0, p_1] \cup \dots \cup [p_{N-1}, p_0]$$

be the polygonal boundary.

Because $\theta_N = 2\pi$, the ideal construction satisfies $p_N = p_0$. Exact closure here is a consequence of reusing the periodic endpoint; it is not a consequence of finite decimal representation of π .

Principal result

Exact circle benchmark. For the regular inscribed N -gon P_N ,

$$d_H(\Gamma_r, P_N) = r \left(1 - \cos \frac{\pi}{N}\right), \quad N_{\text{ins}}(r, \varepsilon) = \left\lceil \frac{\pi}{\arccos(1 - \varepsilon/r)} \right\rceil.$$

Thus the finite sample requirement is exactly determined within this representation family, and asymptotically $N_{\text{ins}}(r, \varepsilon) \asymp \varepsilon^{-1/2}$ for fixed r .

3.2 Chord length and perimeter

Each edge has length

$$\ell_N = 2r \sin \frac{\pi}{N},$$

and each edge subtends central angle $2\pi/N$. The perimeter is

$$L_N = 2Nr \sin \frac{\pi}{N}.$$

Since $\sin x < x$ for $x > 0$,

$$L_N < 2\pi r,$$

and $L_N \rightarrow 2\pi r$ as $N \rightarrow \infty$.

3.3 Exact Hausdorff error

Theorem 3.1 (Exact Hausdorff error of the regular inscribed polygon). *For $N \geq 3$,*

$$d_H(\Gamma_r, P_N) = r \left(1 - \cos \frac{\pi}{N} \right). \quad (2)$$

Proof. By rotational symmetry it is enough to consider one edge. Its endpoints lie on Γ_r , and the supporting line of the edge has distance

$$a_N = r \cos \frac{\pi}{N}$$

from the origin. Every point of the edge lies on that chord, so its radial distance from the origin is at least a_N and at most r . The distance from a boundary point of the polygon to Γ_r is therefore its radial deficit $r - \|x\|$, whose maximum on the edge is attained at its midpoint and equals $r - a_N$.

Conversely, for any point $y \in \Gamma_r$, the radial line from the origin intersects the polygon boundary in the same angular sector. The maximum inward radial deficit occurs at an edge midpoint, by the symmetry of the regular polygon. Hence

$$\sup_{y \in \Gamma_r} \text{dist}(y, P_N) = r - a_N.$$

The reverse directed distance is bounded by the same quantity because every polygon point lies at radial distance between a_N and r . Therefore the Hausdorff distance is exactly $r - a_N$, which is (2). $\square$

3.4 Convergence to the circle

Corollary 3.2 (Hausdorff convergence). *For fixed $r > 0$,*

$$d_H(\Gamma_r, P_N) \longrightarrow 0 \quad (N \rightarrow \infty).$$

More precisely,

$$d_H(\Gamma_r, P_N) = \frac{\pi^2 r}{2N^2} + O(N^{-4}).$$

Proof. The first statement follows immediately from (2). The second follows from

$$1 - \cos x = \frac{x^2}{2} - \frac{x^4}{24} + O(x^6)$$

with $x = \pi/N$. $\square$

Remark 3.3 The convergence statement must not be interpreted as saying that a finite polygon is literally a circle. Every finite P_N remains a polygon. The circle is obtained only as the limiting object in the specified metric.

3.5 Minimum sample count for a prescribed Hausdorff tolerance

Let $0 < \varepsilon < r$. From (2),

$$d_H(\Gamma_r, P_N) \leq \varepsilon$$

if and only if

$$r \left(1 - \cos \frac{\pi}{N} \right) \leq \varepsilon.$$

Hence the least admissible integer within the regular inscribed family is

$$N_{\text{ins}}(r, \varepsilon) = \left\lceil \frac{\pi}{\arccos(1 - \varepsilon/r)} \right\rceil. \quad (3)$$

Theorem 3.4 (Exact minimum within the inscribed regular family). *For $0 < \varepsilon < r$, $N_{\text{ins}}(r, \varepsilon)$ is the least integer $N \geq 3$ for which the regular inscribed N -gon satisfies*

$$d_H(\Gamma_r, P_N) \leq \varepsilon.$$

Proof. The error in (2) is strictly decreasing in N . Solving the inequality gives the continuous threshold in (3); taking the ceiling gives the least admissible integer. $\square$

As $\varepsilon/r \rightarrow 0$,

$$N_{\text{ins}}(r, \varepsilon) \sim \pi \sqrt{\frac{r}{2\varepsilon}}. \quad (4)$$

Thus sample count under this specific representation satisfies

$$N_{\text{ins}}(r, \varepsilon) = \Theta(\varepsilon^{-1/2})$$

for fixed r .

3.6 A refined centered regular-polygon model

The inscribed polygon is a natural and useful representation, but it is not the only regular-polygon choice. Consider a centered regular N -gon with circumradius $R > 0$, allowed to lie either inside or outside Γ_r . Put

$$\alpha := \frac{\pi}{N}.$$

Its inradius is $R \cos \alpha$.

For a centered regular N -gon with circumradius R , the distance from a vertex to Γ_r is $|R - r|$, while the distance from an edge midpoint to Γ_r is $|R \cos \alpha - r|$. Hence

$$d_H(\Gamma_r, P_{N,R}) \geq \max\{|R - r|, |R \cos \alpha - r|\}.$$

The right-hand side is minimized at the point where the two deviations have equal magnitude on opposite sides of r , namely

$$R - r = r - R \cos \alpha,$$

which gives

$$R^* = \frac{2r}{1 + \cos \alpha}, \quad R^* - r = r - R^* \cos \alpha = r \frac{1 - \cos \alpha}{1 + \cos \alpha}.$$

Thus the lower bound becomes

$$E_N^{\text{bal}} = r \frac{1 - \cos \alpha}{1 + \cos \alpha} = r \tan^2 \frac{\pi}{2N}. \quad (5)$$

It remains to show that this lower bound is attained. For the balanced choice, $R^* > r$ and $a := R^* \cos \alpha < r$. The regular polygon is convex and its radial coordinate along each edge ranges from a at the edge midpoint to R^* at the vertices. Consequently, for every $x \in P_{N,R^*}$,

$$\text{dist}(x, \Gamma_r) = ||x|| - r \leq \max\{R^* - r, r - a\} = E_N^{\text{bal}}.$$

For a point $y \in \Gamma_r$, the radial ray from the origin through y meets the corresponding polygon edge at a point whose radial coordinate is at least a . Hence

$$\text{dist}(y, P_{N,R^*}) \leq r - a = E_N^{\text{bal}}.$$

At an edge midpoint $y_m \in \Gamma_r$, equality holds:

$$\text{dist}(y_m, P_{N,R^*}) = r - a = E_N^{\text{bal}}.$$

Therefore

$$d_H(\Gamma_r, P_{N,R^*}) = E_N^{\text{bal}}.$$

Thus R^* is the minimax circumradius within the centered regular N -gon family.

Consequently, the least integer $N \geq 3$ for which this balanced family admits Hausdorff error at most ε is

$$N_{\text{bal}}(r, \varepsilon) = \max \left\{ 3, \left\lceil \frac{\pi}{2 \arctan \sqrt{\varepsilon/r}} \right\rceil \right\}, \quad (6)$$

for $\varepsilon > 0$. As $\varepsilon/r \rightarrow 0$,

$$N_{\text{bal}}(r, \varepsilon) \sim \frac{\pi}{2} \sqrt{\frac{r}{\varepsilon}}.$$

This refinement is not presented as a new best-approximation theorem. Its purpose here is to demonstrate a general methodological point: even after fixing the broad label “regular polygon,” representation choices matter, and an exact optimum within one restricted family need not be globally optimal among all possible polygons.

3.7 Chord-length resolution is a different requirement

If a maximum chord length $s \in (0, 2r)$ is prescribed, then

$$2r \sin \frac{\pi}{N} \leq s$$

if and only if

$$N \geq \frac{\pi}{\arcsin(s/(2r))}.$$

Hence the least admissible polygon order is

$$N_{\text{chord}}(r, s) = \max \left\{ 3, \left\lceil \frac{\pi}{\arcsin(s/(2r))} \right\rceil \right\}. \quad (7)$$

A chord-length specification and a Hausdorff specification are different constraints and should not be conflated.

4 Certified finite-representation pipeline

4.1 Nearest-grid quantization

Let $h\mathbb{Z}^2$ be the square lattice with spacing $h > 0$. Let

$$Q_h : \mathbb{R}^2 \rightarrow h\mathbb{Z}^2$$

be any nearest-grid quantizer with a fixed tie-breaking rule. Then

$$\|Q_h(x) - x\| \leq \frac{h}{\sqrt{2}} \quad (x \in \mathbb{R}^2). \quad (8)$$

Indeed, every point lies in a grid square whose diagonal is $h\sqrt{2}$, and some grid vertex is at most half that diagonal away.

For the ideal polygon, define

$$\tilde{p}_k := Q_h(p_k),$$

and let $\tilde{P}_{N,h}$ be the polygon obtained by joining the quantized vertices cyclically.

Lemma 4.1 (Segment perturbation bound). *Let $a, b, a', b' \in \mathbb{R}^2$. If*

$$\|a - a'\| \leq \eta, \quad \|b - b'\| \leq \eta,$$

then

$$d_H([a, b], [a', b']) \leq \eta.$$

Proof. For $t \in [0, 1]$,

$$\|(1-t)a + tb - ((1-t)a' + tb')\| \leq (1-t)\|a - a'\| + t\|b - b'\| \leq \eta.$$

This gives one directed bound; exchanging primed and unprimed endpoints gives the reverse bound. $\square$

Proposition 4.2 (Grid quantization error). *For nearest-grid vertex rounding,*

$$d_H(P_N, \tilde{P}_{N,h}) \leq \frac{h}{\sqrt{2}}.$$

Proof. Apply the segment perturbation lemma to each corresponding edge and take the maximum over the finitely many edges. $\square$

Theorem 4.3 (Two-stage geometric certificate). *For every $N \geq 3$ and $h > 0$,*

$$d_H(\Gamma_r, \tilde{P}_{N,h}) \leq r \left(1 - \cos \frac{\pi}{N}\right) + \frac{h}{\sqrt{2}}. \quad (9)$$

Proof. The Hausdorff triangle inequality gives

$$d_H(\Gamma_r, \tilde{P}_{N,h}) \leq d_H(\Gamma_r, P_N) + d_H(P_N, \tilde{P}_{N,h}),$$

and the two terms are bounded by the exact polygon theorem and the quantization proposition. $\square$

Remark 4.4 The sum in (9) is a rigorous upper bound. It is not a claim that the two layers attain their individual worst cases simultaneously, nor that the corresponding errors are probabilistically independent.

4.2 A numerical perturbation layer

Let $\hat{P}_{N,h}$ be the computed polygon and suppose there is a verified bound

$$\|\hat{p}_k - \tilde{p}_k\| \leq \eta_{\text{num}} \quad \text{for every } k.$$

The quantity η_{num} may come from a floating-point proof, an interval enclosure, fixed-point analysis, or another certified arithmetic model. It is not automatically equal to a bit count.

Proposition 4.5 (Three-stage geometric certificate). *Under the preceding assumption,*

$$d_H(\Gamma_r, \hat{P}_{N,h}) \leq r \left(1 - \cos \frac{\pi}{N}\right) + \frac{h}{\sqrt{2}} + \eta_{\text{num}}. \quad (10)$$

Proof. The segment perturbation lemma gives

$$d_H(\tilde{P}_{N,h}, \hat{P}_{N,h}) \leq \eta_{\text{num}}.$$

Apply the Hausdorff triangle inequality to the three stages. $\square$

More generally, whenever

$$X \rightarrow D_0 \rightarrow D_1 \rightarrow \cdots \rightarrow D_m$$

has certified transition errors $\delta_0, \dots, \delta_{m-1}$ and the chosen metric is defined on all intermediate objects, one obtains

$$d(X, D_m) \leq \sum_{j=0}^{m-1} \delta_j.$$

This general statement is elementary; its value is that the error sources remain explicit and auditable.

4.3 Structural validity and exact closure

4.3.1 Exact closure by construction

At the ideal level,

$$p_N = p_0.$$

After quantization, exact closure is preserved if the representation explicitly reuses the same stored endpoint.

Proposition 4.6 (Closure by endpoint identity). *If a stored vertex sequence satisfies*

$$q_N := q_0,$$

then the represented polygonal path is exactly closed in the representation space.

Proof. The final and initial vertices are definitionally identical, so their difference is zero. $\square$

4.3.2 Closure under independent perturbation

Suppose instead that q_0 and q_N are computed independently and

$$\|q_0 - p_0\| \leq \eta, \quad \|q_N - p_0\| \leq \eta.$$

Then

$$\|q_N - q_0\| \leq 2\eta. \quad (11)$$

Proof. By the triangle inequality,

$$\|q_N - q_0\| \leq \|q_N - p_0\| + \|q_0 - p_0\| \leq 2\eta.$$

□

Thus small geometric error does not imply exact structural closure. Closure can instead be enforced as a discrete identity.

5 Decision stability and numerical precision

5.1 Decision margins

For $x \notin \mathbb{Z}$, define

$$\Delta_{\mathbb{Z}}(x) := \text{dist}(x, \mathbb{Z}) = \min\{\{x\}, 1 - \{x\}\},$$

where $\{x\}$ is the fractional part.

Lemma 5.1 (Decision stability of the ceiling map). *Let $x \notin \mathbb{Z}$. If*

$$|\hat{x} - x| < \Delta_{\mathbb{Z}}(x),$$

then

$$\lceil \hat{x} \rceil = \lceil x \rceil.$$

Proof. The open interval $(x - \Delta_{\mathbb{Z}}(x), x + \Delta_{\mathbb{Z}}(x))$ contains no integer. Both x and $\hat{x}$ lie in that interval, hence between the same pair of consecutive integers and therefore have the same ceiling. □

The lemma separates value error from decision margin. A computation does not necessarily need a highly accurate value if the final discrete decision is well separated from its decision boundary.

5.2 Application to a circle step count

Suppose a representation uses a prescribed step scale $s > 0$ and chooses

$$N = \left\lceil \frac{2\pi r}{s} \right\rceil.$$

Let

$$x = \frac{2\pi r}{s}, \quad x \notin \mathbb{Z}.$$

If $\hat{\pi}$ is used in place of π , then

$$|\hat{x} - x| = \frac{2r}{s} |\hat{\pi} - \pi|.$$

Therefore the sufficient condition

$$|\hat{\pi} - \pi| < \frac{s}{2r} \Delta_{\mathbb{Z}}\left(\frac{2\pi r}{s}\right) \tag{12}$$

certifies the same integer step count.

The lesson is not that there is a universal number of digits of π . The required precision depends on the representation decision and its decision margin.

5.3 Geometric tolerance and integer decisions

Let

$$\alpha(r, \varepsilon) := \arccos\left(1 - \frac{\varepsilon}{r}\right), \quad 0 < \varepsilon < r.$$

For the inscribed regular model,

$$N_{\text{ins}} = \lceil \pi / \alpha(r, \varepsilon) \rceil.$$

If $\pi / \alpha(r, \varepsilon) \notin \mathbb{Z}$, then a sufficient condition for preserving this integer is

$$|\hat{\pi} - \pi| < \alpha(r, \varepsilon) \Delta_{\mathbb{Z}} \left(\frac{\pi}{\alpha(r, \varepsilon)} \right). \quad (13)$$

The same phenomenon occurs for any discrete decision that is stable away from its boundary.

5.4 What the precision of π actually controls

The motivating question concerned the precision of π for a continuous circle. The analysis shows why that question is representation-dependent.

5.4.1 What irrationality does not imply

The irrationality or transcendence of π does not prevent finite circles or closed polygonal approximations from being represented. A finite polygon closes because its final vertex is identified with its initial vertex. Integer-based raster constructions can avoid evaluating π altogether. Thus

$$\pi \notin \mathbb{Q}$$

is not itself a finite-resource obstruction.

5.4.2 When the precision of π matters

The precision of π matters when a particular representation or discrete decision depends numerically on it: angular coordinates, circumference conversions, arc-length calculations, or formulas that feed into an integer threshold. In those situations, the relevant quantity is a certified uncertainty sufficient to preserve the final representation or decision.

The decision-margin inequalities in Section 5 therefore give a more meaningful interpretation of “how much precision for π ” than a universal digit count. The mathematical object does not impose one intrinsic decimal precision requirement; the chosen representation and task do.

6 Resource-allocation theory

6.1 Separable power-law models

Consider positive continuous resource parameters $q_i > 0$, $i = 1, \dots, m$. Suppose the certified error admits the separable envelope

$$\hat{E}(q) \leq \overline{E}(q) := \sum_{i=1}^m A_i q_i^{-\alpha_i}, \quad A_i > 0, \quad \alpha_i > 0, \quad (14)$$

and suppose the resource cost is

$$C(q) = \sum_{i=1}^m c_i q_i^{\beta_i}, \quad c_i > 0, \quad \beta_i > 0, \quad (15)$$

The resulting continuous *envelope-allocation* problem is

$$\min_{q_i > 0} \sum_{i=1}^m c_i q_i^{\beta_i} \quad \text{subject to} \quad \sum_{i=1}^m A_i q_i^{-\alpha_i} \leq \varepsilon. \quad (16)$$

Set

$$e_i = A_i q_i^{-\alpha_i}, \quad p_i = \frac{\beta_i}{\alpha_i}, \quad K_i = c_i A_i^{\beta_i/\alpha_i}.$$

Then (16) becomes

$$\min_{e_i > 0} \sum_{i=1}^m K_i e_i^{-p_i} \quad \text{subject to} \quad \sum_{i=1}^m e_i \leq \varepsilon. \quad (17)$$

Theorem 6.1 (Unique continuous error allocation). *For $K_i > 0$ and $p_i > 0$, problem (17) has a unique minimizer. The constraint is active, and there is a unique $\lambda > 0$ such that*

$$e_i^* = \left(\frac{p_i K_i}{\lambda} \right)^{1/(p_i+1)}, \quad \sum_{i=1}^m e_i^* = \varepsilon. \quad (18)$$

Proof. Each summand $K_i e_i^{-p_i}$ is strictly convex on $(0, \infty)$ because

$$\frac{d^2}{de_i^2} K_i e_i^{-p_i} = p_i(p_i + 1) K_i e_i^{-p_i-2} > 0.$$

Thus the objective is strictly convex. Since the objective is strictly decreasing in every e_i , an optimum cannot have $\sum_i e_i < \varepsilon$; otherwise increasing some e_i would reduce the objective while maintaining feasibility. Hence the constraint is active.

The Lagrangian is

$$L(e, \lambda) = \sum_i K_i e_i^{-p_i} + \lambda \left(\sum_i e_i - \varepsilon \right).$$

Stationarity yields

$$-p_i K_i e_i^{-p_i-1} + \lambda = 0,$$

which gives (18). The left side of the scalar constraint equation is a strictly decreasing continuous function of λ , tending to $+\infty$ as $\lambda \downarrow 0$ and to 0 as $\lambda \rightarrow \infty$. Hence there is a unique $\lambda > 0$, and strict convexity gives a unique global minimizer. $\square$

Corollary 6.2 (Equal scaling exponents). *If $p_i = p$ for all i , then*

$$e_i^* = \varepsilon \frac{K_i^{1/(p+1)}}{\sum_j K_j^{1/(p+1)}},$$

and the minimum relaxed cost is

$$C^*(\varepsilon) = \left(\sum_{i=1}^m K_i^{1/(p+1)} \right)^{p+1} \varepsilon^{-p}. \quad (19)$$

The theorem optimizes the separable certified envelope $\bar{E}$, not necessarily the exact certificate $\hat{E}$. It is a continuous relaxation and does not assert that rounding q_i^* to integers yields the exact integer optimum.

6.2 Continuous versus discrete optimization

If a resource such as sample count must lie in $\mathbb{N}$, the final problem is an integer optimization problem. In general, the continuous optimum need not round to the exact integer optimum. This distinction is important and will be retained throughout the paper.

6.3 The circle as a resource-allocation model

6.3.1 Two geometric resources

Using

$$1 - \cos x \leq \frac{x^2}{2},$$

the inscribed circle certificate obeys

$$E(N, h) \leq \frac{A}{N^2} + Bh, \quad A := \frac{\pi^2 r}{2}, \quad B := \frac{1}{\sqrt{2}}. \quad (20)$$

Suppose, only as a specified toy model for a fixed-area full grid, that

$$C(N, h) = c_N N + c_h h^{-2}, \quad c_N, c_h > 0.$$

The term h^{-2} is not an intrinsic cost of a circle; it models the number of grid cells in a fixed two-dimensional region.

For fixed $h \in (0, \varepsilon/B)$, the smallest admissible integer $N \geq 3$ satisfying the sufficient certificate is

$$N(h) = \max \left\{ 3, \left\lceil \sqrt{\frac{A}{\varepsilon - Bh}} \right\rceil \right\}. \quad (21)$$

Hence the integer optimization of the *certificate* reduces exactly to the one-dimensional search

$$\min_{0 < h < \varepsilon/B} [c_N N(h) + c_h h^{-2}]. \quad (22)$$

This is an exact reduction of the surrogate certificate problem, not an exact solution of the true geometric optimum over all possible representations.

6.3.2 Unique minimizer of the relaxed toy model

Ignoring the integer ceiling,

$$F(h) = c_N \sqrt{\frac{A}{\varepsilon - Bh}} + c_h h^{-2}.$$

Its derivatives are

$$F'(h) = \frac{c_N \sqrt{A} B}{2(\varepsilon - Bh)^{3/2}} - 2c_h h^{-3}, \quad (23)$$

$$F''(h) = \frac{3c_N \sqrt{A} B^2}{4(\varepsilon - Bh)^{5/2}} + 6c_h h^{-4} > 0. \quad (24)$$

Moreover,

$$F'(h) \rightarrow -\infty \quad (h \downarrow 0), \quad F'(h) \rightarrow +\infty \quad (h \uparrow \varepsilon/B).$$

Therefore:

Proposition 6.3 (Unique optimum of the relaxed two-resource certificate). *The relaxed toy problem has a unique global minimizer $h^* \in (0, \varepsilon/B)$, characterized by*

$$\frac{c_N \sqrt{A} B}{2(\varepsilon - Bh^*)^{3/2}} = 2c_h (h^*)^{-3}. \quad (25)$$

Proof. Strict convexity follows from (24). The derivative changes sign from negative to positive at least once by the endpoint limits, and strict convexity makes the zero unique. The unique stationary point is therefore the unique global minimizer. $\square$

6.3.3 Error budgets

Suppose a representation chain has certified contributions $E_j \leq \varepsilon_j$ and

$$E_{\text{total}} \leq \sum_j E_j.$$

Then any nonnegative allocation satisfying

$$\sum_j \varepsilon_j \leq \varepsilon$$

is sufficient. Equal budgeting,

$$\varepsilon_j = \varepsilon/m,$$

is a choice rather than a theorem. The optimal allocation depends on the error laws and cost sensitivities of the individual layers.

6.3.4 Certified bottlenecks

For the three-layer circle certificate define

$$e_s = r \left(1 - \cos \frac{\pi}{N} \right), \quad e_g = \frac{h}{\sqrt{2}}, \quad e_n = \eta.$$

Then

$$E \leq e_s + e_g + e_n.$$

If a particular resource change affects only e_j , its maximum possible improvement in this certificate is at most e_j . Thus a contribution that is already much smaller than the others has limited immediate leverage under this specific upper bound. This is a rigorous statement about the certificate, not a universal rule that an implementation should always refine the largest term.

6.4 Certificate-compatible representation conversion

The preceding definition measures the cost of one representation scheme in isolation. To compare two schemes rigorously, it is useful to formalize the common situation in which one finite representation can be converted into another. The conversion itself may introduce geometric error and resource cost, and it may preserve or destroy structural validity. These effects should be part of the comparison rather than hidden in an informal statement that one representation is “more compact.”

For a representation scheme $\mathcal{R}$, let $\Phi_{\mathcal{R}}$ denote its realization map from an encoded finite object to the geometric object on which the fidelity metric is evaluated. Thus the actual error of an encoded representation D is measured by $d(X, \Phi_{\mathcal{R}}(D))$. This makes the representation-level framework explicit even when the stored object itself is not a subset of the ambient geometric space.

Definition 6.4 (Certificate-compatible conversion) Let $\mathcal{R}$ and $\mathcal{S}$ be two representation schemes using the same fidelity metric d . A conversion $T : \mathcal{D}_{\mathcal{R}} \rightarrow \mathcal{D}_{\mathcal{S}}$ is called *certificate-compatible* with conversion tolerance $\tau \geq 0$ and conversion cost $\kappa \geq 0$ if, for every representation D admissible under the source contract,

- (i) $P_{\mathcal{R}}(X, D) = 1$ implies $P_{\mathcal{S}}(X, T(D)) = 1$; and
- (ii) the target certificate satisfies

$$\hat{E}_{\mathcal{S}}(X, T(D)) \leq \hat{E}_{\mathcal{R}}(X, D) + \tau;$$

and

(iii) the resource cost of performing and storing the conversion is at most κ .

Condition (ii) is deliberately stated at the level of certified error rather than only at the level of actual metric distance. This is what is needed for a theorem about $\mathcal{C}^{\text{cert}}$ rather than merely about actual feasibility.

Lemma 6.5 (Certificate-preserving complexity transfer). *Suppose $T : \mathcal{D}_{\mathcal{R}} \rightarrow \mathcal{D}_{\mathcal{S}}$ is certificate-compatible with parameters (τ, κ) . Then, for every $\varepsilon > 0$,*

$$\mathcal{C}_{\mathcal{S}, \mathcal{K}_{\mathcal{S}}}^{\text{cert}}(X, \varepsilon + \tau) \leq \mathcal{C}_{\mathcal{R}, \mathcal{K}_{\mathcal{R}}}^{\text{cert}}(X, \varepsilon) + \kappa, \quad (26)$$

whenever the source and target usage contracts are compatible with the stated conversion.

Proof. Take any source representation D satisfying

$$\hat{E}_{\mathcal{R}}(X, D) \leq \varepsilon, \quad P_{\mathcal{K}_{\mathcal{R}}}(X, D) = 1.$$

Certificate compatibility gives

$$\hat{E}_{\mathcal{S}}(X, T(D)) \leq \hat{E}_{\mathcal{R}}(X, D) + \tau \leq \varepsilon + \tau,$$

and structural preservation gives $P_{\mathcal{K}_{\mathcal{S}}}(X, T(D)) = 1$. The conversion cost adds at most κ . Thus every source-feasible certified representation induces a target-feasible certified representation with the stated overhead. Taking the infimum gives (26). $\square$

Corollary 6.6 (Bidirectional comparison). *If there are certificate-compatible conversions from $\mathcal{R}$ to $\mathcal{S}$ with parameters (τ_{RS}, κ_{RS}) and from $\mathcal{S}$ to $\mathcal{R}$ with parameters (τ_{SR}, κ_{SR}) , then the two certified complexity functions bound one another up to the corresponding tolerance and resource overheads. In particular, a representation family cannot be declared intrinsically superior solely from its encoding size if conversion overhead or contract preservation has been ignored.*

The theorem is deliberately elementary at the metric level. Its contribution is not a new form of the triangle inequality; it is the explicit insertion of conversion error, structural-validity preservation, and conversion cost into the certified representation-complexity comparison. This turns representation comparison into a checkable relation rather than a collection of unrelated complexity examples.

The three-stage circle pipeline provides an immediate instance. The conversion from the ideal polygon to the quantized polygon has $\tau = h/\sqrt{2}$, while a verified numerical implementation contributes a further conversion tolerance η_{num} . The existing additive certificate is therefore recovered as an instance of a general certificate-compatible conversion chain. The new point is that the same chain also carries resource overheads and structural predicates, so it can be used to compare different representation families under an explicitly stated contract.

6.5 Representation-dependent resource complexity

6.5.1 Polygonal sample-count complexity

Under the representation contract in which each polygon vertex costs one unit and the regular inscribed family is used, (4) gives

$$\mathcal{C}_{\text{poly}}(\Gamma_r, \varepsilon) \sim \pi \sqrt{\frac{r}{2\varepsilon}}. \quad (27)$$

Thus

$$\mathcal{C}_{\text{poly}}(\Gamma_r, \varepsilon) = \Theta(\varepsilon^{-1/2})$$

for fixed r .

This is a representation-specific complexity law, not an intrinsic complexity of the abstract circle.

6.5.2 Full-grid raster complexity under an explicit contract

Consider a different contract: a full square grid over a fixed region containing the circle is stored, with cell width h . To obtain a sharp complexity law rather than only a sufficient upper bound, assume that the chosen digitization satisfies a two-sided first-order estimate

$$ch \leq d_H(\Gamma_r, \Gamma_{r,h}) \leq Ch \quad (28)$$

for all sufficiently small h , with constants $0 < c \leq C$ independent of h . The precise constants and geometric or topological hypotheses depend on the digitization model [6].

The upper bound gives a sufficient choice $h \leq \varepsilon/C$, while the lower bound implies that any representation satisfying $d_H(\Gamma_r, \Gamma_{r,h}) \leq \varepsilon$ must have

$$h \leq \varepsilon/c.$$

If the full representation stores a fixed-area two-dimensional window, then its number of cells is

$$\Theta(h^{-2}).$$

Consequently,

$$\mathcal{C}_{\text{grid,full}}(\Gamma_r, \varepsilon) = \Theta(\varepsilon^{-2}) \quad (\varepsilon \rightarrow 0), \quad (29)$$

under this two-sided first-order geometric assumption and the stated full-grid storage contract. A one-sided error certificate alone would justify only the corresponding $O(\varepsilon^{-2})$ upper bound. A sparse boundary representation has a different cost law, and an implicit representation has another.

6.5.3 Finite-bit coordinates for polygonal samples

As a joint precision model, suppose a polygon uses $N \geq 3$ vertices and each coordinate is represented with an integer number $b \geq 1$ of fractional bits. Assume the certified end-to-end geometric bound has the form

$$E_{\text{cert}}(N, b) \leq \frac{A}{N^2} + D2^{-b}, \quad A, D > 0, \quad (30)$$

where A represents the geometric discretization coefficient and D absorbs the coordinate range and quantizer constant. Assume the normalized storage cost is

$$C(N, b) = Nb. \quad (31)$$

For a concrete implementation with a fixed number of coordinate fields per vertex, the true storage differs only by a constant factor, so the asymptotic conclusions below are unchanged.

It is important to state exactly what is being optimized. The quantity

$$C_{\text{mix}}^*(\varepsilon) := \min_{\substack{N \in \mathbb{N}, N \geq 3 \\ b \in \mathbb{N}, b \geq 1}} \left\{ Nb : \frac{A}{N^2} + D2^{-b} \leq \varepsilon \right\}$$

is the minimum cost of the *stated certificate model*. It is not the unconstrained geometric optimum over all possible representations.

Theorem 6.7 (Sharp mixed discretization–finite-bit allocation). *For fixed $A, D > 0$, let*

$$L_\varepsilon := \ln \frac{D}{\varepsilon}.$$

As $\varepsilon \rightarrow 0$,

$$C_{\text{mix}}^*(\varepsilon) = \frac{\sqrt{A/\varepsilon}}{\ln 2} [L_\varepsilon + \ln L_\varepsilon + O(1)]. \quad (32)$$

More precisely, the continuous relaxation of (6.5.3) has a unique minimizer $(\bar{N}_\varepsilon, \bar{b}_\varepsilon)$, characterized by

$$\delta_\varepsilon := \frac{D2^{-\bar{b}_\varepsilon}}{\varepsilon} \in (0, 1)$$

and the scalar equation

$$\delta_\varepsilon (L_\varepsilon - \ln \delta_\varepsilon) = 2(1 - \delta_\varepsilon). \quad (33)$$

Furthermore,

$$\delta_\varepsilon = \frac{2}{L_\varepsilon} + O\left(\frac{\ln L_\varepsilon}{L_\varepsilon^2}\right), \quad (34)$$

$$\bar{N}_\varepsilon = \sqrt{\frac{A}{\varepsilon}} \left[1 + \frac{1}{L_\varepsilon} + O\left(\frac{\ln L_\varepsilon}{L_\varepsilon^2}\right)\right], \quad (35)$$

$$\bar{b}_\varepsilon = \frac{L_\varepsilon + \ln L_\varepsilon - \ln 2 + o(1)}{\ln 2}, \quad (36)$$

and the continuous relaxed cost is

$$\bar{C}_{\text{mix}}(\varepsilon) = \frac{\sqrt{A/\varepsilon}}{\ln 2} [L_\varepsilon + \ln L_\varepsilon + 1 - \ln 2 + o(1)]. \quad (37)$$

The integer optimum has the same leading and $\ln L_\varepsilon$ terms as in (32); the unavoidable integer rounding contributes only an $O(1)$ term inside the square brackets.

Proof. First consider the continuous relaxation $N > 0, b > 0$. At an optimum the certificate constraint is active. Introduce

$$\delta = \frac{D2^{-b}}{\varepsilon}, \quad 0 < \delta < 1.$$

Then

$$N = \sqrt{\frac{A}{\varepsilon(1-\delta)}}, \quad b = \frac{L_\varepsilon - \ln \delta}{\ln 2}.$$

Hence the relaxed objective is

$$\bar{C}(\delta) = \frac{\sqrt{A/\varepsilon}}{\ln 2} \frac{L_\varepsilon - \ln \delta}{\sqrt{1-\delta}}.$$

The derivative of the dimensionless factor is

$$\frac{d}{d\delta} \left[\frac{L_\varepsilon - \ln \delta}{\sqrt{1-\delta}} \right] = \frac{\delta(L_\varepsilon - \ln \delta) - 2(1-\delta)}{2\delta(1-\delta)^{3/2}}.$$

Define

$$\psi_\varepsilon(\delta) := \delta(L_\varepsilon - \ln \delta) - 2(1-\delta).$$

Then

$$\psi'_\varepsilon(\delta) = L_\varepsilon + 1 - \ln \delta > 0$$

for $0 < \delta < 1$ and sufficiently small ε , while $\psi_\varepsilon(\delta) \rightarrow -2$ as $\delta \downarrow 0$ and $\psi_\varepsilon(1) = L_\varepsilon > 0$. Thus there is a unique root, proving (33) and uniqueness of the relaxed minimizer.

To obtain the asymptotics, rewrite the optimality equation as

$$\delta_\varepsilon (L_\varepsilon - \ln \delta_\varepsilon + 2) = 2.$$

The unique solution satisfies

$$\delta_\varepsilon = \frac{2}{L_\varepsilon} + O\left(\frac{\ln L_\varepsilon}{L_\varepsilon^2}\right),$$

which follows by one substitution into the displayed equation; a sharper expansion is

$$\delta_\varepsilon = \frac{2}{L_\varepsilon} - \frac{2(\ln L_\varepsilon - \ln 2 + 2)}{L_\varepsilon^2} + O\left(\frac{(\ln L_\varepsilon)^2}{L_\varepsilon^3}\right).$$

Using

$$(1 - \delta_\varepsilon)^{-1/2} = 1 + \frac{\delta_\varepsilon}{2} + O(\delta_\varepsilon^2)$$

gives (35). Also,

$$-\ln \delta_\varepsilon = \ln L_\varepsilon - \ln 2 + o(1),$$

which gives (36); substituting both expansions into the relaxed objective yields (37).

Finally, the integer optimum is bounded below by the continuous relaxation. For an upper bound, round $\bar{N}_\varepsilon$ and $\bar{b}_\varepsilon$ upward to integers. The rounded pair remains certificate-feasible, and the additional product cost is

$$O(\bar{N}_\varepsilon + \bar{b}_\varepsilon + 1) = O\left(\sqrt{\frac{1}{\varepsilon}} + \ln \frac{1}{\varepsilon}\right),$$

which is

$$\frac{\sqrt{A/\varepsilon}}{\ln 2} O(1)$$

relative to the bracketed scale in (32). Thus integer rounding can alter the bounded $O(1)$ term but cannot change the L_ε or $\ln L_\varepsilon$ coefficients. This proves (32). $\square$

Corollary 6.8 (Asymptotically vanishing quantization share). *In the continuous relaxation, the optimal fraction of the total certified error budget assigned to finite-bit quantization is*

$$\frac{D2^{-\bar{b}_\varepsilon}}{\varepsilon} = \delta_\varepsilon \sim \frac{2}{\ln(D/\varepsilon)}.$$

Thus the optimal allocation does not split the two error sources equally. As the target tolerance becomes more stringent, the quantization share tends to zero logarithmically, while the geometric discretization receives the remaining $1 - \delta_\varepsilon$ fraction.

The logarithmic correction in (32) is the useful feature of this explicit model. A coarse separate-constraint argument gives only the weaker $\Theta(\varepsilon^{-1/2} \log(1/\varepsilon))$ scaling. The optimization above resolves the coupling between the two error mechanisms and identifies the asymptotically optimal continuous error allocation. The result is deliberately stated for the explicit certificate model (30); it is not a claim about all polygonal compression or all mixed-precision algorithms. Joint polygonal approximation and vertex quantization are established problems in vector-data compression [4], and mixed-precision numerical analysis likewise studies how discretization and quantization errors should be balanced [9, 10]. The present contribution is the closed-form asymptotic analysis of this particular geometric certificate and storage contract.

6.5.4 Representation competition

The preceding examples illustrate the central conceptual point:

$$\varepsilon^{-1/2}, \quad \varepsilon^{-2}, \quad \varepsilon^{-1/2} \log(1/\varepsilon)$$

can arise from different representation or cost models for the same continuous object. Thus there is no representation-independent sentence of the form “a circle requires N resources.” The mathematically meaningful quantity is

$$\mathcal{C}_{\mathcal{R}, \mathcal{K}}(\Gamma_r, \varepsilon),$$

with the representation scheme, usage contract, tolerance, and cost model all stated.

6.6 Further consequences and feasibility

6.6.1 Monotonicity

For fixed r ,

$$E_N = r \left(1 - \cos \frac{\pi}{N} \right)$$

is strictly decreasing in $N \geq 3$. For fixed N and the nearest-grid certificate,

$$E(N, h, 0) = E_N + \frac{h}{\sqrt{2}}$$

is strictly increasing in h . These monotonicities justify one-dimensional feasibility searches when all other parameters are fixed.

6.6.2 Physical-resolution feasibility window

Suppose the physical sample system imposes a minimum arclength separation δ between consecutive marks and the regular inscribed representation is required to meet Hausdorff tolerance ε . Then necessarily

$$N \leq \left\lfloor \frac{2\pi r}{\delta} \right\rfloor$$

while the geometric tolerance requires

$$N \geq N_{\text{ins}}(r, \varepsilon).$$

Therefore a necessary feasibility condition is

$$N_{\text{ins}}(r, \varepsilon) \leq \left\lfloor \frac{2\pi r}{\delta} \right\rfloor. \quad (38)$$

This statement is deliberately about a physical-resolution contract. It does not alter the mathematical fact that the ideal circle contains infinitely many points.

6.6.3 A representation-complexity transfer principle

Suppose a representation family has integer or continuous resource parameter $m \rightarrow \infty$ and, for sufficiently large m , the error and cost satisfy

$$E_{\mathcal{R}}(m) \sim A_{\mathcal{R}} m^{-\alpha_{\mathcal{R}}}, \quad C_{\mathcal{R}}(m) \sim B_{\mathcal{R}} m^{\beta_{\mathcal{R}}},$$

with $A_{\mathcal{R}}, B_{\mathcal{R}}, \alpha_{\mathcal{R}}, \beta_{\mathcal{R}} > 0$. Assume also that $E_{\mathcal{R}}$ is eventually decreasing and $C_{\mathcal{R}}$ is eventually increasing. Then solving $E_{\mathcal{R}}(m) \leq \varepsilon$ gives

$$C_{\mathcal{R}}(\varepsilon) \sim B_{\mathcal{R}} A_{\mathcal{R}}^{\beta_{\mathcal{R}}/\alpha_{\mathcal{R}}} \varepsilon^{-\beta_{\mathcal{R}}/\alpha_{\mathcal{R}}}.$$

Thus the approximation order of a representation induces a corresponding resource-complexity order once the cost law is specified. This is a bookkeeping principle rather than a novelty claim, but it provides a compact language for comparing representations. If only two-sided order bounds are known, the corresponding conclusion is a Θ -law without the displayed asymptotic constant.

7 Purpose-specific adequacy

7.1 Why geometric accuracy need not be the final specification

For many applications, geometric distance is an intermediate condition rather than the final objective. A visual display, measurement system, control computation, or engineering analysis may care about a task performance rather than about reconstruction error itself. Task-based quantization makes this distinction explicit in signal-processing settings [14]; rate–distortion–perception theory similarly separates numerical distortion from perceptual quality [13].

The present paper does not attempt to construct a universal human utility function. Instead, a task is represented abstractly.

Definition 7.1 (Task adequacy functional) Let T denote a task or specification, and let

$$U_T(X, D)$$

be a scalar performance functional comparing the use of representation D with the ideal object X . A threshold $U_{\min}$ defines task adequacy when

$$U_T(X, D) \geq U_{\min}.$$

This functional may represent a measured performance quantity, a certified engineering quantity, a detection margin, a perceptual criterion, or another explicitly defined task metric.

7.2 A transfer theorem from geometric error to task adequacy

Theorem 7.2 (Lipschitz transfer of task adequacy). *Suppose there exists $L_T \geq 0$ such that for every admissible representation D ,*

$$|U_T(X, D) - U_T(X, X)| \leq L_T d(X, D).$$

Assume the ideal object has a positive task margin

$$\Delta_T := U_T(X, X) - U_{\min} > 0.$$

Then every representation satisfying

$$d(X, D) < \frac{\Delta_T}{L_T}$$

when $L_T > 0$ is task-adequate:

$$U_T(X, D) > U_{\min}.$$

If $L_T = 0$, the task performance is representation-independent under the stated model.

Proof. By (7.2),

$$U_T(X, D) \geq U_T(X, X) - L_T d(X, D).$$

If $d(X, D) < \Delta_T/L_T$, then

$$U_T(X, D) > U_T(X, X) - \Delta_T = U_{\min}.$$

The case $L_T = 0$ is immediate. $\square$

This theorem makes the “enough for the purpose” intuition mathematically precise without asserting a universal human threshold. If a particular application can establish or estimate L_T and Δ_T , then a task requirement can be converted into a sufficient geometric tolerance, after which the representation-complexity problem can be solved.

7.3 Diminishing returns

The circle already contains a geometric diminishing-returns law:

$$E_N \sim \frac{\pi^2 r}{2N^2}.$$

Doubling N reduces the leading-order error by a factor of approximately four. A separate task functional may exhibit additional saturation, but such a statement is application-dependent and is not assumed universally in this paper.

The mathematically safe conclusion is therefore:

The marginal value of additional representation resources must be assessed relative to the error law, the task sensitivity, and the chosen cost model; there is no representation-independent universal “enough.”

7.4 Static versus multiresolution representations

The main analysis assumes a fixed representation scale. If the viewing scale changes, as in zoomable graphics or multiresolution visualization, a single static sample count may cease to be adequate. The natural extension is then a scale-indexed family $\{\mathcal{R}_j\}$ with an error requirement ε_j at each scale. This paper does not develop the dynamic problem further.

8 Literature context and status of the contribution

8.1 Polygonal approximation and quantization

Imai and Iri formulated minimum-error and minimum-complexity polygonal approximation problems [1]; Chin and Chan developed algorithmic results for related formulations [2]; Kolesnikov and Fránti treated closed discrete curves [3]. Kolesnikov also studied polygonal approximation together with vertex quantization in vector-data encoding, including minimum distortion at a given bit rate and minimum bit rate at a given maximum deviation [4]. These results show that joint approximation and quantization are established problems.

The present paper therefore does not present its general resource-allocation template as a new compression theory. Its purpose is to expose the representation pipeline and keep geometric error, quantization, arithmetic perturbation, structural validity, and task specifications as auditable layers.

8.2 Digital geometry

Klette and Rosenfeld provide a broad treatment of digital geometry [5]. Lachaud and Thibert prove geometric and topological properties of Gauss digitizations for broad classes of regular sets [6]. The present paper deliberately uses simpler nearest-grid vertex quantization in its main certificate and does not attempt a complete theory of digital topology or digitized manifolds.

8.3 Robust predicates and precision

Shewchuk’s adaptive-precision framework demonstrates that arithmetic precision should be driven by the uncertainty relevant to the discrete geometric decision rather than by a fixed global digit count [7]. The decision-margin formulation here is compatible with that principle but is stated at a representation level rather than as a replacement for robust-predicate algorithms.

8.4 Mixed precision and resource allocation

McCormick, Benzaken, and Tamstorf analyze algebraic rounding error in mixed-precision multi-grid methods, while Tamstorf, Benzaken, and McCormick connect quantization and discretization error and develop progressive precision [9, 10]. Their results establish that increasing discretization and holding precision fixed can eventually expose quantization error. The present paper imports the conceptual separation of error sources into a geometric setting but does not reproduce their numerical-analysis theory.

Zinkhahn, Leube, de Barros, and Nowak study computational resource allocation across spatial, temporal, and stochastic discretization dimensions for a specified prediction task [12]. This is a close conceptual precedent for the general optimization viewpoint. The present paper specializes the object of study to deterministic geometric representations and emphasizes representation and structural validity.

8.5 Perception and task-aware representation

Blau and Michaeli formalize the rate–distortion–perception trade-off, showing that distortion and perceptual quality are distinct constraints [13]. Shlezinger, Eldar, and Rodrigues develop hardware-limited task-based quantization in which the quantizer is designed in light of the downstream task [14]. These works make clear that “enough” representation is already an established research question in information theory and signal processing.

Accordingly, the present paper does not claim novelty for purpose-aware representation in general. Its narrower focus is the deterministic geometric case with explicit representation contracts, structural constraints, and certified geometric/error bounds. The conversion relation in Section 6.4 is supporting framework machinery rather than the principal novelty claim. The principal quantitative contribution is the sharp mixed discretization–finite-bit allocation result in Section 6.5: for the explicit certificate $A/N^2 + D2^{-b} \leq \varepsilon$ and normalized storage cost Nb , the analysis identifies the $\ln \ln(1/\varepsilon)$ correction and the asymptotically optimal continuous error share $\delta_\varepsilon \sim 2/\ln(D/\varepsilon)$.

8.6 Limitations and claim boundaries

8.6.1 The quantization model is intentionally simple

Nearest-grid vertex rounding is not the same as Gauss digitization of a solid or boundary. The bound $h/\sqrt{2}$ is rigorous for the stated vertex-quantization model but does not encode the full topology of digital shapes.

8.6.2 The numerical perturbation parameter is abstract

The parameter η_{num} is a certified geometric perturbation. Translating a hardware format or an IEEE floating-point precision into such a bound requires a separate rounding-error analysis for the actual algorithm.

8.6.3 The cost model is not universal

Memory, time, energy, communication, latency, hardware area, and monetary cost are different resources. They may be scalarized only after a weighting or application-specific objective is specified. Multiobjective Pareto analysis is an alternative when no defensible scalarization exists.

8.6.4 Certified complexity may exceed actual complexity

Triangle-inequality certificates are often conservative. A certificate-feasible resource level need not be truly optimal for the actual error because worst-case mechanisms may not occur simultaneously.

8.6.5 Representation-class dependence

The quantity $\mathcal{C}_{\mathfrak{R},\mathcal{K}}$ depends on the chosen representation class and usage contract. Enlarging the class can lower the infimum, possibly dramatically. There is therefore no universal representation-independent “finite complexity” of a continuous object in the present framework.

8.6.6 Task adequacy requires task information

The theorem in Section 7 converts task stability and a task margin into a geometric tolerance. It does not supply L_T or Δ_T . Those quantities must come from the actual application or an independently justified task model. In particular, the paper does not claim a universal visual threshold for human observers.

8.7 Research status and contribution statement

The manuscript should be read as a mathematically controlled framework and solvable model rather than as a claim that polygonal approximation, digital geometry, task-based quantization, rate-distortion theory, or adaptive precision were invented here.

The certificate-preserving conversion lemma is supporting machinery. The principal quantitative contribution claimed here is instead the sharp asymptotic analysis of the explicit mixed discretization-finite-bit certificate in Section 6.5. The theorem goes beyond the coarse $\Theta(\varepsilon^{-1/2} \log(1/\varepsilon))$ order by identifying the $\ln \ln(1/\varepsilon)$ correction, deriving the unique continuous optimizer through a scalar equation, and showing that integer rounding preserves the first two asymptotic terms.

The claim is intentionally narrow. Kolesnikov’s work already studies polygonal approximation jointly with vertex quantization and bit-rate constraints [4], while mixed-precision numerical analysis studies the balancing of discretization and quantization errors [9, 10]. The present manuscript does not claim to establish joint approximation-and-quantization as a new research problem. Instead, it contributes an explicit, closed-form asymptotic solution for the particular circle-derived error law $A/N^2 + D2^{-b}$ under the stated Nb storage contract. The balanced regular-polygon calculation is also treated as established background; Wintraecken and Vegter analyze the corresponding optimally rescaled regular-polygon problem under Hausdorff distance [15].

Because novelty is necessarily a literature-dependent judgment, the manuscript uses the formulation “to the best of our knowledge” for this narrow asymptotic result rather than

claiming priority over all related resource-allocation work.

9 Open problems and extensions

- O1. Sharp quantization geometry.** Determine the exact worst-case Hausdorff error of nearest-grid rounding for circle-polygon vertices as a function of r , N , h , and grid orientation, rather than using the generic $h/\sqrt{2}$ certificate.
- O2. Representation-class lower bounds.** Establish nontrivial lower bounds for specific families of finite geometric representations under common usage contracts, rather than only deriving upper-bound certificates.
- O3. Cross-representation optimality.** Compare polygonal, sparse-grid, full-raster, and implicit integer representations under explicitly equivalent task and access contracts.
- O4. Finite-precision complexity.** Replace the abstract perturbation η by a verified bit-level floating-point model for a concrete circle-generation algorithm and determine the exact asymptotic cost of certified representation.
- O5. Task stability.** Derive useful bounds for concrete task functionals, including measurement, geometric predicates, or perceptual models, so that task margins can be translated into representation tolerances.
- O6. Multiresolution representation.** Extend the static analysis to a family of representations indexed by viewing scale or spatial frequency, with explicit update and storage costs.
- O7. Curvature-adaptive sampling.** For general C^2 or smoother curves, derive resource-optimal adaptive sampling laws under a specified geometric certificate and compare them with uniform sampling.
- O8. Topology and structural validity.** Extend the Boolean structural predicate to connectness, non-self-intersection, orientation preservation, and topology under digitization.
- O9. Multiobjective costs.** Replace a scalar cost by a Pareto set when memory, energy, latency, and arithmetic complexity cannot be defensibly combined into one number.
- O10. Higher-dimensional objects.** Extend the framework to spheres and smooth surfaces, where curvature, anisotropy, mesh topology, and quantization interact more strongly.

10 Conclusion

The central lesson is that a continuous geometric object and a finite representation of it should be treated as different mathematical objects connected by an explicit representation contract.

For the circle, a regular inscribed polygon satisfies

$$d_H(\Gamma_r, P_N) = r \left(1 - \cos \frac{\pi}{N} \right),$$

so its required number of samples for tolerance ε is known exactly within that representation family and scales as $\Theta(\varepsilon^{-1/2})$. The sequence converges to the circle in Hausdorff distance, but no finite polygon is literally the circle. A finite physical system is additionally constrained by its own resolution, and a simple minimum-separation assumption gives a finite upper bound on the number of distinguishable physical samples.

Spatial quantization and finite numerical perturbation can be added without collapsing logically distinct mechanisms. The resulting certificates are rigorous upper bounds, while structural requirements such as exact closure remain separate Boolean conditions. A discrete decision can require far less numerical precision than would be needed to approximate its underlying real quantity to a fixed number of digits, provided the decision margin is certified.

The resource-allocation formulation shows why “make the resolution arbitrarily high” is not a mathematically complete prescription. Increasing one resource may reduce one error term while another resource becomes dominant, and the optimal allocation depends on the cost law. The general power-law theorem gives a clean continuous model for that trade-off; the circle specialization exhibits the exact discrete constraints that must be handled afterward.

For the explicit mixed certificate $A/N^2 + D2^{-b} \leq \varepsilon$ with normalized storage cost Nb , the paper obtains the sharper asymptotic law

$$C_{\text{mix}}^*(\varepsilon) = \frac{\sqrt{A/\varepsilon}}{\ln 2} \left[\ln \frac{D}{\varepsilon} + \ln \ln \frac{D}{\varepsilon} + O(1) \right].$$

The continuous relaxation further shows that the optimal fraction of the error budget spent on bit quantization is asymptotically $2/\ln(D/\varepsilon)$ rather than a fixed split. This result is a property of the stated certificate and storage contract, not a universal law for all finite representations.

Finally, the same circle can admit representations with fundamentally different resource-accuracy laws. Thus the meaningful question is not the intrinsic finite resource requirement of an abstract circle, but the minimum resource required by a specified representation, usage contract, tolerance, and task.

The methodological principle can therefore be summarized as

Separate the object, representation, errors, structural requirements, and costs; certify before optimizing.

A Elementary inequalities and asymptotics

For all real x ,

$$1 - \cos x \leq \frac{x^2}{2}.$$

For $x \rightarrow 0$,

$$1 - \cos x = \frac{x^2}{2} - \frac{x^4}{24} + O(x^6).$$

Also,

$$\sin x = x + O(x^3), \quad \tan x = x + O(x^3).$$

These expansions give the stated N^{-2} approximation laws for the inscribed and balanced regular-polygon models.

B Numerical illustration

Take $r = 1$ and $\varepsilon = 10^{-2}$. For the inscribed regular model,

$$N_{\text{ins}} = \left\lceil \frac{\pi}{\arccos(0.99)} \right\rceil = 23.$$

The exact polygonal error is

$$1 - \cos \frac{\pi}{23} \approx 9.3141 \times 10^{-3}.$$

If nearest-grid vertex rounding with $h = 10^{-3}$ is added, the certified quantization contribution is

$$\frac{h}{\sqrt{2}} \approx 7.071 \times 10^{-4}.$$

The resulting end-to-end certificate is therefore

$$9.3141 \times 10^{-3} + 7.071 \times 10^{-4} \approx 1.0021 \times 10^{-2},$$

which exceeds the target 10^{-2} . Thus the jointly certified pair $(N, h) = (23, 10^{-3})$ is infeasible under the additive certificate, whereas $N = 24$ gives

$$1 - \cos \frac{\pi}{24} + \frac{10^{-3}}{\sqrt{2}} \approx 9.2622 \times 10^{-3},$$

and is feasible in this particular model. The example is deliberately simple: its purpose is to show why independently choosing a polygonal resolution and a grid resolution can violate a common end-to-end tolerance.

For a numerical check of the mixed discretization–finite-bit theorem, take $r = 1$, so $A = \pi^2/2$, and set $D = 1$ and $\varepsilon = 10^{-12}$. The continuous relaxation gives

$$L_\varepsilon \approx 27.6310, \quad \delta_\varepsilon \approx 0.06170, \quad \bar{N}_\varepsilon \approx 2.2933 \times 10^6, \quad \bar{b}_\varepsilon \approx 43.8818.$$

Thus about 6.17% of the certified error budget is assigned to finite-bit quantization in the continuous model. The nearby integer pair

$$(N, b) = (2,287,405, 44)$$

satisfies

$$\frac{A}{N^2} + 2^{-b} \approx 9.999996 \times 10^{-13} < 10^{-12},$$

with normalized cost $Nb = 100,645,820$. This numerical check is included only to illustrate the asymptotic allocation; the theorem above is the analytic result.

C Methodological scope

The paper intentionally follows a strict labeling discipline. A result proved from the stated assumptions is called a theorem, proposition, lemma, or corollary. A statement that depends on an explicit model is labeled as such. Practical advice that is not universal is described as an operational rule or interpretation rather than a theorem. Literature precedents are identified as established mathematics rather than absorbed into the contribution. Open questions are stated as open problems rather than presented as settled results.

References

- [1] H. Imai and M. Iri, Polygonal approximations of a curve—formulations and algorithms, in *Machine Intelligence and Pattern Recognition*, Vol. 6, pp. 71–86, Elsevier, 1988. DOI: [10.1016/B978-0-444-70467-2.50011-4](https://doi.org/10.1016/B978-0-444-70467-2.50011-4).

- [2] F. Y. L. Chin and W. S. Chan, Approximation of polygonal curves with minimum number of line segments or minimum error, *International Journal of Computational Geometry and Applications*, 6(1):59–77, 1996. DOI: [10.1142/S0218195996000058](https://doi.org/10.1142/S0218195996000058).
- [3] A. Kolesnikov and P. Fránti, Polygonal approximation of closed discrete curves, *Pattern Recognition*, 40(4):1282–1293, 2007. DOI: [10.1016/j.patcog.2006.09.002](https://doi.org/10.1016/j.patcog.2006.09.002).
- [4] A. Kolesnikov, Optimal encoding of vector data with polygonal approximation and vertex quantization, in *Image Analysis*, 14th Scandinavian Conference, SCIA 2005, Lecture Notes in Computer Science, Springer, 2005. DOI: [10.1007/11499145_120](https://doi.org/10.1007/11499145_120).
- [5] R. Klette and A. Rosenfeld, *Digital Geometry: Geometric Methods for Digital Picture Analysis*, Morgan Kaufmann, San Francisco, 2004.
- [6] J.-O. Lachaud and B. Thibert, Properties of Gauss digitized shapes and digital surface integration, *Journal of Mathematical Imaging and Vision*, 54(2):162–180, 2016. DOI: [10.1007/s10851-015-0595-7](https://doi.org/10.1007/s10851-015-0595-7).
- [7] J. R. Shewchuk, Adaptive precision floating-point arithmetic and fast robust geometric predicates, *Discrete & Computational Geometry*, 18(3):305–363, 1997. DOI: [10.1007/PL00009321](https://doi.org/10.1007/PL00009321).
- [8] J. E. Bresenham, A linear algorithm for incremental digital display of circular arcs, *Communications of the ACM*, 20(2):100–106, 1977. DOI: [10.1145/359423.359432](https://doi.org/10.1145/359423.359432).
- [9] S. F. McCormick, J. Benzaken, and R. Tamstorf, Algebraic error analysis for mixed-precision multigrid solvers, *SIAM Journal on Scientific Computing*, 43(5):S392–S419, 2021. DOI: [10.1137/20M1348571](https://doi.org/10.1137/20M1348571).
- [10] R. Tamstorf, J. Benzaken, and S. F. McCormick, Discretization-error-accurate mixed-precision multigrid solvers, *SIAM Journal on Scientific Computing*, 43(5):S420–S447, 2021. DOI: [10.1137/20M1349230](https://doi.org/10.1137/20M1349230).
- [11] B. Yang, X. Wang, F. W. B. Li, B. Xie, X. Liang, and Z. Jiang, 3D mesh compression and transmission for mobile robotic applications, *International Journal of Advanced Robotic Systems*, 13, 2016. DOI: [10.5772/62035](https://doi.org/10.5772/62035).
- [12] P. C. Leube, F. P. J. de Barros, W. Nowak, and R. Rajagopal, Towards optimal allocation of computer resources: Trade-offs between uncertainty quantification, discretization and model reduction, *Environmental Modelling & Software*, 50:97–107, 2013. DOI: [10.1016/j.envsoft.2013.08.008](https://doi.org/10.1016/j.envsoft.2013.08.008).
- [13] Y. Blau and T. Michaeli, Rethinking lossy compression: The rate-distortion-perception tradeoff, in *Proceedings of the 36th International Conference on Machine Learning*, PMLR, Vol. 97, pp. 675–685, 2019.
- [14] N. Shlezinger, Y. C. Eldar, and M. R. D. Rodrigues, Hardware-limited task-based quantization, *IEEE Transactions on Signal Processing*, 67(20):5223–5238, 2019. DOI: [10.1109/TSP.2019.2935864](https://doi.org/10.1109/TSP.2019.2935864).
- [15] M. H. M. J. Wintraecken and G. Vegter, On the optimal triangulation of convex hypersurfaces, whose vertices lie in ambient space, *Mathematics in Computer Science*, 9(3):345–353, 2015. DOI: [10.1007/s11786-014-0216-7](https://doi.org/10.1007/s11786-014-0216-7).
- [16] H. Imai and M. Iri, Computational-geometric methods for polygonal approximations of a curve, *Computer Vision, Graphics, and Image Processing*, 36(1):31–41, 1986. DOI: [10.1016/S0734-189X\(86\)80027-5](https://doi.org/10.1016/S0734-189X(86)80027-5).